

Temporal Associative Memory

Online sequence learning

The “temporal” associative memory component learns higher-order sequences and next-token prediction on the fly.

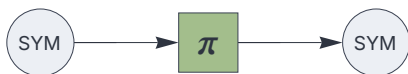
Requirements

```
<< "Circuits.m"
```

Circuit

```
{
  "$schema": "https://creatingintelligence.org/schemas/circuit-v1.json",
  "hyperparameters": {"default": [3000, 10]},
  "dataflow": [
    {"component": "input", "plugin": "codec", "send": [11]},
    {"component": "temporal", "plugin": "permutation", "send": [21], "receive": [11], "capacity": 8,
    "decimate": 0.8},
    {"component": "output", "plugin": "codec", "receive": [21]}
  ]
}
```

```
circ = Circuit[json];
c = circ["function"];
circ["schematics"][ImageSize→220]
```



Dataset

Generate a 2000 character random string.

```
chars = CharacterRange["A", "Z"]
```

```
{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z}
```

```
dataset = RandomChoice[ chars, 2000];
```

```
StringJoin[dataset]
```

```
HWACYFUCSQMFVJQZFWBXCXGEMPRTVSXZRMFSSZXGYNDCACBWQRIWPDOSRCOLQUINZYJXK:  
FRNZFYRGOOZAGOEWAQMVEKBTFWPBIRHGONGLNQZERAUNLNXFCSUMRGIMEWPQSDIVMFQ:  
DVEUFTYFYEQJPVXVPLUXHCTHRTSMTUXQFGORISENMNEJATTUJDXRYRHQKIGXRYROMDAPS:  
JWWENFYAADOBJANGKSFVDIIIMOFTAVJPTTUZVOPRFXYESRCBGMCEQJBFTEAHHVZKZKRU:  
JIIIIHIKKECROFAAEOWFIDQPGTLXLYSBBXMTHEADGZWGVVAVMPLKLUWVEZQPRRBBWKGWE:  
KPOVIZCRAVNXCJELZYQTFSHCEYIQUQGDSFIKKBLIMBJQOWJKJAJIZWQEPZFDUUZAXSYXJ:  
BWVVSNMCEHNHIEALITNPTXDBKPEPNFYIDVTQADQVRPEHHZPDHVTHTJNQPIEFFGBBDOMEL:  
ZCQCBDDJJSMDDFAWQQLNAJYPHRVCMWFNRLSCKNIBWNSXGEIZBWTBCCHPXFXUHGYYQHKREE:  
UJXVSLNNRSTOSULRBOIXVZVPRMCPFPQPNUDNEMKPHBOCFGHMRYPFBHADLLPBBTGUSKBFR:  
QVKEXMCPDJPJGJXTGEDZUIFRABIPKMLNHAUTWPYTEMPIOJYPQFKSVAWMZVXFWODRJNVNM:  
WAMEORPPZPHJFRMXJUJUMHETXQTPRNCXHNHOEALCOGJMRXRGHZRAFYSJGESUIZDCXZRK:  
PGTMFEMMAKREWNHNWODWRATSWWRMTZHXFLDXEEKDGKINYHQTNEOFHVNKGKZOAFCAUGBB:  
CZXNLAGHPEAIEDQELXIHYZQPGYZQRFUIYJEYFHKMSDYJQCBYOUXFNIPDCAZXCWTSYTYX:  
NTEVMCRLSOWBNJSEAJLIOBAYPCBUFNVMJUYJXYKOYCYJNBUSNDBRCLIIQIKWZAKVOLDUE:  
SHBTIBVVKUAVGRCIORSGQTWSHPLGJFOMCMMTHZUAYMAYPPAEXCHWJXBFKMDAWHGKBWFJY:  
EVAUIRTUHDPRRTNXDGTJWKQLFMXFXPFOBFTSUFUTWKKOWDUIXSISSKLCVBJVJNWRKFMQ:  
DMUFGMYHGMGYIGJZYWBUDQZGHFZKAEBPQEHLGQUIHWRXZHJESPHQMAPIUKJZPBONENGO:  
MTIRYYQCXUUYMGAZOHGRZTYIANKOVUHPCSGAVNXWVDUIDKFIOUSXHAXJKLUSFFSWLKUUQ:  
MJUDEACHTKWWROUQMHBVBNRPDLVZRAYHAZPDRAWZYGQBPJJIJAYQIVEECBCPKTWGGKFC:  
ZMORPLSESANFBTYSBPZQPQREAEGUQLODBKASPJYIWMNOCWVNHMBZHQWMSQEDLSUJDYD:  
JYDDYGOYUOPXDBIZRAGXSYVQDUYZSRMKHSQLPEZZTEQPDVKYINKCNHWQZMHZCKVOINX:  
ZRBSTDMBNOMUXFHDENGUYJXUCYMSNFRDNRKLJMBYLSZFZJQNGWUUEOQDVHERQUTJWAA:  
UFGSWGZRIFYJASKDQUNRPXWJHBVGQPEISFKIQKBVUYTQUKWUVIHJEJBKCDLZRFGGKGT:  
UONLRLWEFLWFXQDTRFNZPQPTQNKAVMXKEZAPZZCJDVIAEQWQHGJGRYBBDACONKHTAHCW:  
ZPDSQZDXASCRAWOHCIMUAUVEEJAHELOXQBZGXXBJVOVABYPROTKRJONAZXFIOQHZWAK:  
DSHODIOLXQWHZGWZSLMRNACHYRVPSSFELHXUMLYJGRTXAHWBRICJFFAXEDBCQGCTSZGJZ:  
YSEVIZPVLABDJMRXQJKPNDSXCKOYHPIMXMBQRSXSEKCZYRXOAQLRQFNCFSZPFDLMYTTC:  
KLBORCIAWFYYRDCIWFMHKUXYEPFRFYUPBGGEUBCJCCIAQLQZVCTUANBNMKWDFDXPOSBM:  
MAJFICEZGBOTNMHDZVTPRLBYXVCCPWBCUZACXENMPISHOPLLRFRGMNINVGCZTPGT
```

Prediction score

```
Do[
  predictions = c /@ dataset;
  Print["    iteration = ", i,
    ", prediction score = ", N[Mean@Boole@MapThread[SameQ,
      {dataset, RotateRight[predictions]}]]],
    {i, 1, 5}];
```

iteration = 1, prediction score = 0.033

iteration = 2, prediction score = 0.989

iteration = 3, prediction score = 1.

iteration = 4, prediction score = 1.

iteration = 5, prediction score = 1.